Sahaj Bhadja

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Summary

Associate Software Engineer with 2+ years of experience building **Python/TypeScript systems**, **React applications**, secure APIs, cloud services, and **reliable data-intensive products** across production environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Reduced dashboard latency from 1.2s to 350ms through **React interfaces, SQL indexing, and query optimization**, supporting 500+ users with responsive, maintainable analytics workflows across isolated tenant environments.
- Achieved 99%+ reliability across 5K+ monthly jobs through **validation, reconciliation, and secure error handling**, cloud storage, CI/CD releases, and production monitoring for dependable data-intensive software workflows at scale.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, secure ingestion, AWS RDS, and CI/CD workflows.
- Processed 1K+ files per batch using **concurrent Python services**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during sustained production traffic and model execution reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **maintainable microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments under sustained traffic.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **React and Node.js platform**, designing reusable interfaces for customers, billing, allocation, permissions, and notifications with real-time updates across distributed locations reliably.
- Reduced resource wastage by 42% through **real-time data synchronization**, implementing JWT access controls, WebSocket updates, Azure monitoring, automated tests, and Docker/Jenkins deployments across eight modular service domains.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Improved cross-platform compatibility by 15% through **React Native and TypeScript interfaces**, integrating Android, iOS, and web booking workflows while reducing design errors by 25% through reusable components.
- Delivered **sub-200ms customer workflows** across 50K+ booking records, integrating pricing, coupons, payment webhooks, Firebase synchronization, and status updates while preventing conflicting reservations across concurrent user sessions reliably.

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, Bash
Frontend: React, React Native, Next.js, HTML, CSS, Tailwind CSS, Responsive Design
Backend & APIs: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, Microservices, JSON, JWT, Caching, Pagination
Databases: PostgreSQL, MySQL, MongoDB, Firebase, AWS RDS, Azure SQL, NoSQL, Data Modeling, Schema Design, Query Optimization
Cloud & Delivery: AWS, GCP, Azure, Docker, Kubernetes, Jenkins, Git, GitHub, CI/CD, Maven
Quality & Security: Unit Testing, Integration Testing, Automated Testing, PyTest, Mockito, Jest, Selenium, Debugging, Monitoring, Error Handling, Fault Isolation, Secure Coding, Role-Based Access Control, Production Reliability

Projects

HMIS - Hospital Management Information System | Live
- Built a **TypeScript healthcare platform** integrating consent-based ABDM APIs, Twilio alerts, and Sentry monitoring, enabling 100+ daily appointments, 2x-faster patient-history access, and 30% lower administrative workload securely.

Big-Brain | Live
- Built a **secure AI document platform** with PostgreSQL, semantic search, role-based access, and real-time collaboration, supporting 50K+ documents while reducing document-analysis time by 60% through optimized retrieval workflows.

FinNexus | Live
- Built a **TypeScript production platform** supporting 50K+ users and 1M+ transactions with real-time synchronization, advanced filtering, MongoDB-backed access, token authorization, Kubernetes deployment, and responsive interfaces reliably at scale.